

CUSTOMER MILESTONE BRIEF

MS #10 - Audit Ledger and Evidence Control Tower

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.10.0
Date	2026-08-11
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #10 - Audit Ledger and Evidence Control Tower converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Makes modernization decisions, evidence, approvals, exceptions, and release posture tamper-evident and reviewable.

Why this matters to a customer

Makes modernization decisions, evidence, approvals, exceptions, and release posture tamper-evident and reviewable.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a versioned, append-only audit contract with actors, roles, actions, subjects, evidence references, visibility, timestamps, sequence numbers, previous-event hashes, and event identities.
- Added deterministic ledger snapshots bound to the canonical knowledge-graph identity, plus validation that detects mutation, deletion, reordering, duplication, broken chains, stale projections, invalid decision hashes, statistics drift, and checkpoint tampering.
- Added deterministic development-readiness, mainframe-equivalence, and release-promotion policy decisions. Planner and builder roles cannot record acceptance decisions, and missing z/OS proof blocks release promotion.
- Added governed exceptions requiring a human approver, named owner, substantive justification, expiry, and compensating controls; mainframe equivalence and release promotion are non-overridable.
- Added optional HMAC-SHA256 checkpoint signing through an environment-only key, wrong-key detection, and explicit unsigned-canonical warnings.
- Added machine-readable and human-readable release evidence dossiers with policy rationale, unresolved gaps, evidence inventory, and ledger checkpoint identity.

- Added a read-only Evidence Control Tower to the graph explorer with promotion posture, policy decisions, audit timeline, dossier, and checkpoint views.
- Added cross-platform audit build and deterministic verification launchers, six portable JSON Schema contracts, tamper/privacy/policy/signature/API/UI tests, and full-suite integration.

How it differs from earlier work

MS #10 builds on MS #9 (z/OSMF Adapter and Connection Simulator). The earlier milestone established its own bounded capability; this milestone adds audit ledger and evidence control tower while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use audit ledger and evidence control tower as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.10.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity